FREEMARKET · SENIOR DATA ENGINEER TAKE-HOME

A trustworthy, GBP-normalised money-flow network

Bronze → Silver → Gold in a single DuckDB file: from four raw sources to a graph-ready Gold dataset, with every number traceable back to an untouched source row.

12,982
transactions · Jul–Dec 2025

£26.15bn
GBP-normalised volume

1,376
network nodes

4,110
directed edges

101
tests gating the build

THE PROBLEM

Commercial and Finance can't answer a basic question

“Who is moving how much money with whom, and when? And what revenue did we earn on it?”

WHY IT'S HARD

- Four raw extracts from different systems, **not built for analytics**: a 4-sheet Excel workbook and three differently-nested JSON files
- Amounts in **many currencies**; FX rates arrive as ~161k unsorted, time-versioned tuples
- The corporate hierarchy (group → legal entity → counterparty) is **implicit across sources**

WHAT WAS ASKED

- A **Medallion pipeline in DuckDB**: one local file, SQL + Python
- A Gold dataset driving a **directed money-flow network**: GBP volume, count and fee revenue per relationship
- **Sliceable** by month/year, **drillable** up and down the group ↔ company hierarchy

ARCHITECTURE

Three layers, six schemas, one file, built to the team's protocol

BRONZE · LAND FAITHFULLY

`raw`
one table per month, source names & types untouched

`live`
months reunited; facts live here once, names conformed

→

SILVER · CLEAN & CONVERT

`core`
JSON unpicked; FX rate *selected* per fact (as-of match)

`shape`
FX *applied* → GBP facts; unpriceable rows quarantined

→

GOLD · MODEL & PRESENT

`data_mart`
entity dimension + money-flow aggregate, row-level provenance

`curated`
graph-ready nodes + edges; reads only from data_mart

THE TOPOLOGY DECISION: CATALOG VS SCHEMA

DuckDB shares the `catalog.schema.table` namespace with Databricks, but a DuckDB catalog is a **physical file**, so the "catalog per layer" idiom would need three attached files and break the single-file constraint. So layers map to **schemas**; if the rule were lifted, they lift mechanically to catalogs (`ATTACH 'bronze.duckdb'`) with schema names unchanged.

THE UNDERPINNING PRINCIPLE

Keep the raw fact pure. Facts land once in Bronze `live` ; FX attaches as a separate keyed table, never as columns bolted onto the fact. Lineage back to an untouched source row is what makes row and measure counts checkable at every layer boundary.

THE CENTRAL MODELLING CHALLENGE

FX: every amount priced at its own instant, or not at all

- **Point-in-time as-of join:** each amount converts at the rate whose validity window spans its settlement instant (`Tx Date + Tx Time` ; fees at their date). Cross-currency totals only compare if each amount is priced at its own moment
- **One pure, isolated unit** (`src/fx.py`): sorts the unsorted rate points once, range-matches `validFrom ≤ t < validTill` by bisect; independently tested with neither DuckDB nor the 18 MB file
- **GBP is an explicit identity** (rate 1.0; it has no series in the file), not a fudge
- **Full lineage:** every priced fact carries the `fx_rate_id` of the exact rate point that priced it

NEVER SILENTLY MIS-PRICED

Anything unpriceable is **quarantined with a stated reason**: kept and auditable, never dropped or null-filled:

`fx_out_of_coverage`	instant outside the file's window
`fx_unknown_currency`	no series for the currency
`fx_no_rate_point`	gap in the series at that instant
`fx_missing_currency`	row has no currency
`fx_missing_instant`	row has no settlement instant

Splitting “which rate” (`core`) from “apply the rate” (`shape`) separates the two ways FX can go wrong into two inspectable stages.

DATA QUALITY

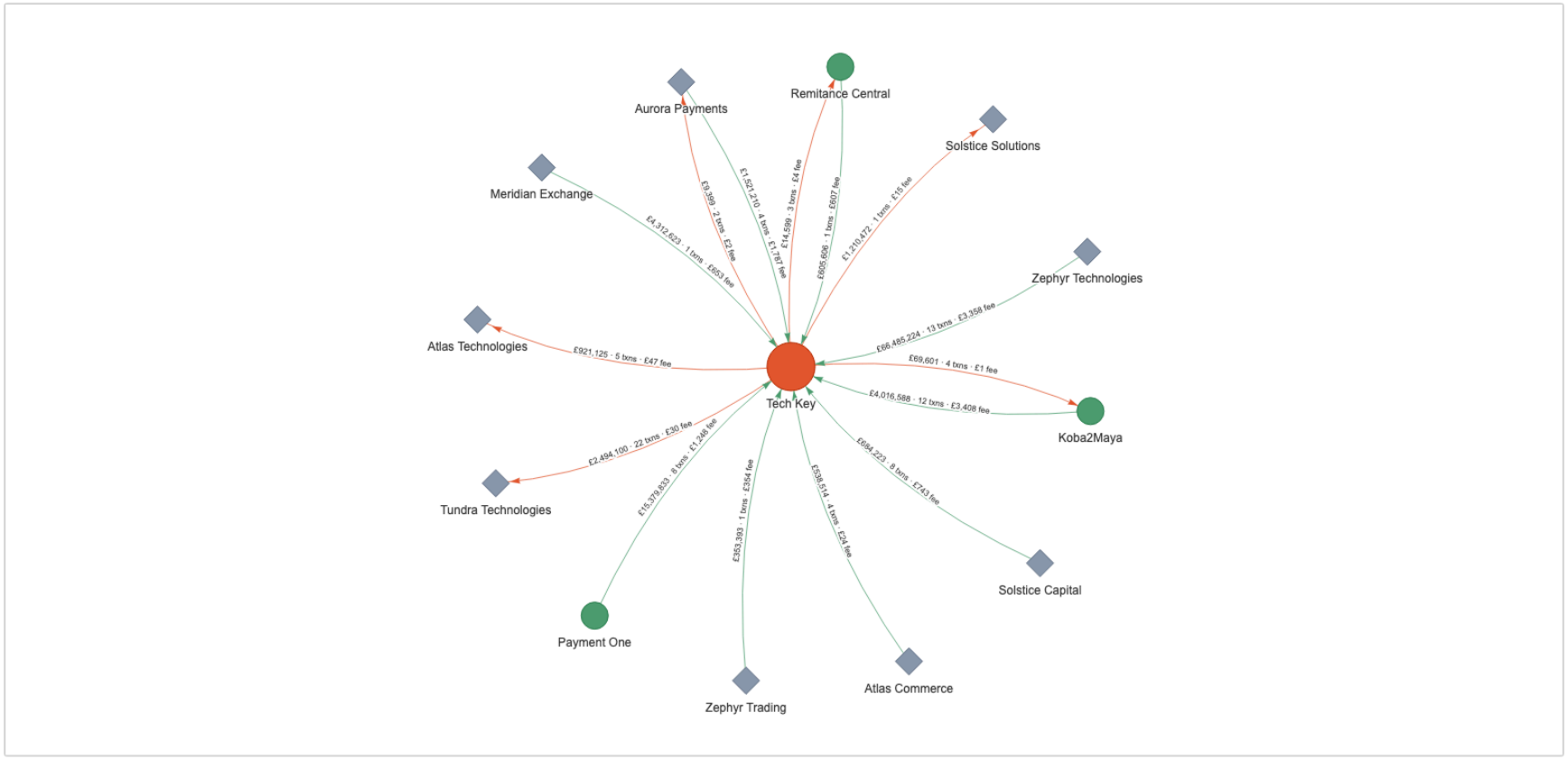
Nothing vanishes silently: every exclusion classified, every boundary gated

WHAT THE DATA THREW UP	COUNT	VERDICT	WHY
Fees with no currency	42	**quarantined**	unpriceable (`fx_missing_currency`); this is the entire quarantine ledger
Transaction IDs in both Deposit & Withdrawals	306	**kept**	distinct streams reusing IDs; fees key on `(link_id, fee_type)` so nothing fans out
Counterparties with no group/company key	1,363	**kept**	standalone by design → diamond nodes, exactly as the brief expects
Counterparties with a key	222	**kept**	all resolve deterministically to a group with zero orphans; no name-matching needed
Duplicate fact key within a stream	0	**would fail loud**	unique key asserted before pricing; a dup would inflate volume and fan out fees
Null amounts · out-of-period rows	0	**quarantine · fail loud**	null amount → quarantined like a bad currency; out-of-period → Bronze raises. Enforced, not just observed, so the next drop can't slip a bad row through

Conservation gates the build: per-month rows sum to `live` · promoted + quarantined = input · the 12,982 transactions land in `curated.edge` uncounted-out · GBP totals unchanged Silver → Gold. Any breach turns `make test` red.

THE DELIVERABLE

Gold `curated` drives the graph directly, with zero reshaping



`make render` → this star map, read straight from `curated.node` + `curated.edge`: focal group **Tech Key**, top-12 counterparts by volume, full period. ● circles = groups (group↔group flow is first-class: 449 such edges) ◆ diamonds = standalone counterparties · directed edges labelled **£ volume · txn count · fee revenue** (green in, orange out).

SLICE & DRILL

Any group, any period, any level of the hierarchy, one `WHERE` clause away

- Every edge carries `month` and `year`. The month grain is additive, so year and full-period views are plain **roll-ups by summation**
- Every edge carries both `focal_group_id` and `focal_company_id` (+ its name). The fact is stored at the **finest company grain**, so the group view is the sum of its companies, and the drill is **recoverable, not lost**
- Proven in SQL (`notebooks/drill.ipynb`): star view → named legal entities → one company's own counterparts → **roll-up reconciles to the penny**

THE DRILL, CONCRETELY (KOBA2MAYA)

Group grain: who does Koba2Maya, as a whole, move money with?
↓ **drill:** which of its 18 legal entities transacted: *two separate entities both named* “Operations Ltd” (£66m and £18m), “Group Holdings £6.7m”…
↓ **drill:** who did the £66m entity itself deal with?
↑ **roll up:** the group total equals the sum of its companies; *drill is lossless*.

Hierarchy comes from deterministic keys only (`dc_id → company → parentGroup`), so there's no name-matching and no guessing. Names duplicate across jurisdictions; the **id is the key**, the name is just the label.

IF THIS WERE A LIVE STREAM

The design already splits at the right seams: four concrete deltas

1. **Swap the ingestion source.** `build_raw_monthly` reads a sheet today; replace that one function with a stream consumer appending events to the Bronze `raw` month partition. Nothing downstream of `live` needs to know.
2. **Make Bronze writes append-only and idempotent.** `INSERT … ON CONFLICT DO NOTHING` on the natural key, so at-least-once delivery and replays can't double-count.
3. **Process incrementally.** Watermark the new rows; `money_flow` is additive on its grain, so a micro-batch is an upsert into the affected month's aggregate, not a full rebuild.
4. **Refresh FX as a live dimension.** A late-arriving rate point re-prices rows quarantined as `fx_no_rate_point`, so nothing stays stuck.

Unchanged: the FX unit, the JSON unpicking, the network model, the quarantine rules, the conservation checks. The transformation logic never cared whether a row came from a sheet or a stream.

KEY DECISIONS

The load-bearing choices, and the alternatives they beat

- **Attach FX, don't bolt it on.** The match lives in its own table keyed to the fact. Rejected: `fx_*` columns on the fact (mutates the raw fact, collapses two failure modes into one).
- **Quarantine is terminal.** Gold reads only promoted facts; no wrong or null number can reach the product. Rejected: null-filled GBP (silently poisons totals).
- **Money-flow at the finest grain.** Company-level rows, group view as roll-up. Rejected: group grain (drill would be unrecoverable, since `curated` reads only `data_mart`).
- **Enforce integrity, don't assume it.** Duplicate keys fail the build, null amounts quarantine; no-ops on this drop, protection for the next one.
- **Deterministic counterpart→group resolution** (`group_id`, else `dc_id → company → group`): zero orphans found, so no fragile name-matching anywhere.
- **One documented deviation** from the build protocol: singular table names (`live.deposit`, not `deposits`), keeping one naming convention across all six schemas; the protocol's substance (one consolidated table per stream) is unchanged.

WHY YOU CAN TRUST IT, AND WHAT'S NEXT

Reproducible by construction, honest about the one unknown

TRUST

- `make pipeline` rebuilds the committed warehouse from scratch, deterministically: same totals every run
- **101 tests** gate the build: the pure FX unit, per-layer invariants, an end-to-end conservation spine, and a write-up completeness tripwire
- Row-level provenance (`source` lists) and FX lineage (`fx_rate_id`) on every Gold row

THE ONE STAKEHOLDER QUESTION

The reference snapshot is from a **different data drop**: its figures don't exist in this data, so it guided *shape*, not numbers. Whether any snapshot figure is a validated reference for a named group and period is the single question worth an email; everything else was resolvable from the data.

NEXT

- Incremental micro-batch ingestion + orchestrator (Dagster/Airflow)
- A Gold DQ-reporting view: the quarantine ledger as a product
- Click-to-expand drill UI on top of `curated`